Probability Distributions

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Part I. Review

Discrete Distributions

- **Geometric Distribution**: Models the number of trials until the first success; X = trial count, θ = success probability per trial.
  - PMF: $P_X(k) = (1-\theta)^{k-1}\theta,\ k \geq 1$
  - MGF: $m_X(t) = \frac{\theta e^t}{1-(1-\theta)e^t}$
  - Moments: $\mathbb{E}(X) = \frac{1}{\theta},\ \operatorname{Var}(X) = \frac{1-\theta}{\theta^2}$
- **Binomial Distribution**: Counts successes in n independent Bernoulli trials; X = success count, n = number of trials, θ = success probability per trial.
  - PMF: $P_X(x) = \binom{n}{x}\theta^x(1-\theta)^{n-x},\ x = 0, 1, \ldots, n$
  - MGF: $m_X(t) = \left(1-\theta+\theta e^t\right)^n$
  - Moments: $\mathbb{E}(X) = n\theta,\ \operatorname{Var}(X) = n\theta(1-\theta)$
- **Negative Binomial Distribution**: Trial number of the r-th success in independent Bernoulli trials ($r = 1$ gives geometric); X = trial index, r = target success count, θ = success probability per trial.
  - PMF: $P_X(x) = \binom{x-1}{r-1}\theta^r(1-\theta)^{x-r},\ x = r, r+1, r+2, \ldots$
  - MGF: $m_X(t) = \frac{\left(\theta e^t\right)^r}{\left(1-(1-\theta)e^t\right)^r}$
  - Moments: $\mathbb{E}(X) = \frac{r}{\theta},\ \operatorname{Var}(X) = \frac{r(1-\theta)}{\theta^2}$
  - Equivalent form: $Y = X - r$ (failures before the r-th success) has PMF $\binom{r+y-1}{y}(1-\theta)^y\theta^r$, $y = 0, 1, 2, \ldots$
- **Hypergeometric Distribution**: Draws r items without replacement from n total items, of which n_1 are of a specified type; X = number of that type in the sample.
  - PMF: $P_X(k) = \frac{\binom{n_1}{k}\binom{n-n_1}{r-k}}{\binom{n}{r}}$
  - Moments: $\mathbb{E}(X) = \frac{rn_1}{n},\ \operatorname{Var}(X) = \frac{rn_1(n-n_1)(n-r)}{n^2(n-1)}$
  - Derived via both direct summation and indicator variable decomposition.
- **Poisson Distribution**: Number of times an event occurs in a fixed window when each occurrence is rare and independent (e.g. rare wins, counts in a short time slice); X = occurrence count, λ = average count in that window ($\mathbb{E}(X) = \operatorname{Var}(X) = \lambda$).

- PMF: $P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!},\ k = 0, 1, 2, \ldots$
- MGF: $m_X(t) = e^{\lambda(e^t - 1)}$
- Moments: $\mathbb{E}(X) = \mathrm{Var}(X) = \lambda$
- Reproductive property: sum of independent Poissons is Poisson.

- **Poisson Process**: Events arrive randomly over time at average rate λ per unit time; N_t = total arrivals by time t; counts in non-overlapping time intervals are independent, and at most one arrival at a time.
  - Derived via differential equations for $P_k(t) = \Pr(N_t = k)$
  - Key result: $N_{t+s} - N_t \sim \mathrm{Poisson}(\lambda s)$
  - Inter-arrival times are Exponential. If the process has rate λ (events per unit time), each gap is $\mathrm{Exponential}(\beta = 1/\lambda)$: equivalently, rate $1/\beta = \lambda$.

Continuous Distributions

- CDF (Cumulative Distribution Function) $F_X(x) = \int_{-\infty}^{x} f_X(y)dy$.
- PDF (Probability Density Function) $f_X(x) = F'_X(x)$
- Expectation: $\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)dx, \quad \mathbb{E}(\phi(X)) = \int_{-\infty}^{\infty} \phi(x) f_X(x)dx$
- MGF (Moment Generating Function) for continuous X: $m_X(t) = \mathbb{E}(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f_X(x)dx$
- Variance: $\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$
- **Gamma Function & Gamma Distribution**: Waiting-time model for the sum of α i.i.d. Exponential(β) stages; X = positive value, α = shape (number of stages), β = scale.
  - Gamma function: $\Gamma(\alpha) = \int_0^\infty z^{\alpha-1}e^{-z}dz$, with properties $\Gamma(\alpha+1) = \alpha\Gamma(\alpha),\ \Gamma(n+1) = n!,\ \Gamma(1/2) = \sqrt{\pi}$
  - Gamma(α, β) PDF: $f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta},\ x > 0$
  - Moments: $\mathbb{E}(X) = \alpha\beta,\ \ \mathrm{Var}(X) = \alpha\beta^2$

- **Exponential Distribution**: Waiting time until the next event in a Poisson process (memoryless); X = waiting time, β = scale (mean waiting time); rate $1/\beta$ (special case $\alpha = 1$ of Gamma).
  - PDF: $f_X(x) = \frac{1}{\beta}e^{-x/\beta},\ x > 0$
  - CDF: $F_X(x) = 1 - e^{-x/\beta}$
  - Poisson link: for a Poisson process of rate λ, inter-arrival time has scale $\beta = 1/\lambda$ (same law as $\mathrm{Exp}(\lambda)$ in rate notation).

- **Chi-Squared Distribution** (χ^2_k): Sum of squares of k independent $N(0,1)$ variables; X = nonnegative value, k = degrees of freedom (special case of Gamma with $\alpha = k/2$, $\beta = 2$).
  - Special case of Gamma with $\alpha = k/2,\ \beta = 2$
  - Moments: $\mathbb{E}(X) = k,\ \ \mathrm{Var}(X) = 2k$

- **Uniform Distribution**: Every value on an interval is equally likely; X = value on $[a, b]$, a and b = interval endpoints.
- **Normal (Gaussian) Distribution**: Bell-shaped model for measurements, sample means, and aggregated noise; X = value, μ = mean, σ^2 = variance.
  - PDF: $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$
  - MGF: $m_X(t) = e^{\mu t + \sigma^2 t^2/2}$
  - Moments: $\mathbb{E}(X) = \mu,\ \ \mathrm{Var}(X) = \sigma^2$

- Standardization: if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$
- Standard normal CDF denoted $\Phi(z)$; empirical rule: ~68% within 1σ, ~95% within 2σ, ~99.7% within 3σ.

- **Weibull Distribution**: Time to failure when the hazard rate is a power of t (reliability / wear-out modeling); X = failure time, α = scale, β = shape ($\beta = 1$ gives Exponential).
  - PDF: $f_X(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta},\ x > 0$
  - Reliability: $R(t) = \Pr(X > t) = e^{-\alpha t^\beta}$; CDF $F_X(t) = 1 - R(t)$
  - Hazard rate: $\rho(t) = \dfrac{f_X(t)}{R(t)} = \beta\alpha t^{\beta-1}$ (constant when $\beta = 1$; increasing when $\beta > 1$)
  - Mean: $\mathbb{E}[X] = \alpha^{-1/\beta}\Gamma\left(1 + \dfrac{1}{\beta}\right)$; calibrate α from a target mean via this formula
  - Reduces to Exponential when $\beta = 1$; memoryless iff $\beta = 1$

Part II. Exercises

EX1. A lot contains $n = 200$ chips. Each chip is independently defective with probability $\theta = 0.03$. Let X = number of defective chips in the lot.

(i) Find $\Pr(3 \le X \le 9)$ using three methods: Exact binomial PMF, Poisson approximation, Normal approximation with continuity correction. Report numeric values relative errors.

(ii) We design package number. Find the minimum lot size n such that we have more than 90% probability that at least 100 chips are good, using the poisson approximation.

(i) $X \sim \text{Binomial}(n = 200,\ \theta = 0.03)$, $\lambda = \mu = 6$, $\sigma^2 = 5.82$, $\sigma \approx 2.4125$.

Exact binomial:
$\Pr(3 \le X \le 9) = \sum_{k=3}^{9} \binom{200}{k} \theta^k (1-\theta)^{200-k} \approx 0.8599$

Poisson:
$\Pr(3 \le X \le 9) \approx \sum_{k=3}^{9} \frac{e^{-6} 6^k}{k!} \approx 0.8541$
Error: 0.0058.

Normal:
$\Pr(3 \le X \le 9) \approx \Phi\left(\frac{9.5-6}{\sigma}\right) - \Phi\left(\frac{2.5-6}{\sigma}\right) \approx \Phi(1.45) - \Phi(-1.45) \approx 0.8532$
Error: 0.0068.

Poisson is slightly closer than Normal here because θ is small.

(ii) Need $\Pr(X \le n - 100) \ge 0.90$ where $X \approx \text{Poisson}(\lambda = 0.03n)$.

At $n = 104$: $\lambda = 3.12$, allow at most 4 defectives,
$\Pr(X \le 4) = \sum_{k=0}^{4} \frac{e^{-3.12} 3.12^k}{k!} \approx 0.795.$

At $n = 105$: $\lambda = 3.15$, allow at most 5 defectives,
$\Pr(X \le 5) = \sum_{k=0}^{5} \frac{e^{-3.15} 3.15^k}{k!} \approx 0.9002.$

Minimum lot size: $n = 105$.

EX2. A help desk receives phone calls. On average there are 4 calls per hour, and calls arrive independently over time (a Poisson process). Let N_t count how many calls have arrived by time t (measured in hours).

(i) What is the probability of at least 2 calls in the first 30 minutes?

(ii) Let W_1 be the waiting time until the first call. Prove the memoryless property.

(iii) Using part (ii), find the probability of exactly 3 calls during the first hour and exactly 2 calls during the second hour.

(iv) Prove by convolution that the sum of two independent Poisson random variables is Poisson.

(v) Using parts (ii) and (iv), are "exactly 4 calls in the first hour" and "exactly 3 calls between the 30-minute mark and the 90-minute mark" independent?

(i) Half an hour is $t = 0.5$, rate $\lambda = 4$, so $N_{0.5} \sim \text{Poisson}(\lambda t) = \text{Poisson}(2)$.

$\Pr\left(N_{0.5} \geq 2\right) = 1 - e^{-\lambda t} - (\lambda t)e^{-\lambda t} = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2} \approx 0.594$

(ii) $W_1 \sim \text{Exponential}(\lambda)$, so $\Pr\left(W_1 > x\right) = e^{-\lambda x}$.

$\Pr\left(W_1 > s + t \mid W_1 > s\right) = \frac{\Pr(W_1 > s+t)}{\Pr(W_1 > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \Pr\left(W_1 > t\right)$

(iii) Independent.

$\Pr\left(N_1 = 3,\ N_2 - N_1 = 2\right) = \frac{e^{-4}4^3}{3!} \cdot \frac{e^{-4}4^2}{2!} = \frac{256}{3}e^{-8} \approx 0.0286$

(iv) Let $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ be independent. By convolution,

$\Pr\left(X + Y = n\right) = \sum_{k=0}^{n} \frac{e^{-\lambda_1}\lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{n!} \sum_{k=0}^{n} \binom{n}{k} \lambda_1^k \lambda_2^{n-k} = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}$

So $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

(v) By (ii), split $[0, 1.5]$ into three disjoint 30-minute intervals with independent counts

$$U = N_{0.5}, \qquad V = N_1 - N_{0.5}, \qquad W = N_{1.5} - N_1,$$

each $\text{Poisson}(2)$. Define the events

$$A = \{N_1 = 4\} = \{U + V = 4\}, \qquad B = \{N_{1.5} - N_{0.5} = 3\} = \{V + W = 3\}.$$

Joint probability $\Pr\left(A \cap B\right) = \Pr\left(U + V = 4,\ V + W = 3\right)$. The two sums share V, so we condition on how many calls fall in the middle half-hour:

$$\Pr\left(A \cap B\right) = \sum_{v=0}^{3} \Pr\left(U = 4 - v,\ V = v,\ W = 3 - v\right).$$

Because U, V, W are independent, each triple factorizes:

$$\Pr(A\cap B)=\sum_{v=0}^{3}\Pr(U=4-v)\,\Pr(V=v)\,\Pr(W=3-v)$$

$$=\sum_{v=0}^{3}\frac{e^{-2}2^{4-v}}{(4-v)!}\cdot\frac{e^{-2}2^{v}}{v!}\cdot\frac{e^{-2}2^{3-v}}{(3-v)!}=e^{-6}\left(\frac{8}{9}+\frac{32}{6}+8+\frac{16}{6}\right)=\frac{152}{9}e^{-6}\approx 0.0419.$$

Marginal probabilities with each interval one hour of rate 4:

$$\Pr(A)=\Pr(N_1=4)=\frac{e^{-4}4^4}{4!}=\frac{32}{3}e^{-4},\qquad \Pr(B)=\Pr(N_{1.5}-N_{0.5}=3)=\frac{e^{-4}4^3}{3!}=\frac{32}{3}e^{-4}.$$

Not independent:

$$\Pr(A)\Pr(B)=\left(\frac{32}{3}e^{-4}\right)^2=\frac{1024}{9}e^{-8}\approx 0.0382\neq\frac{152}{9}e^{-6}\approx 0.0419=\Pr(A\cap B).$$

Both counts are built from V, the middle half-hour. Knowing $N_1=4$ fixes how many calls $U+V$ contributed to the first hour, which changes the distribution of V and hence of $V+W$.

EX3. A fair coin is tossed repeatedly.

(i) What is the probability that the 5^{th} head occurs on the 10^{th} toss?

(ii) Suppose the 10^{th} head occurs on the 25^{th} toss. On which toss does the 5^{th} head most likely occur? At what probability? What about the 8^{th}?

(iii) Following (ii), can we simply use $25\cdot\frac{5}{10}=12.5$ and $25\cdot\frac{8}{10}=20$? Why? Give a rigorous account of the apparant linearity.

(i) This follows Negative Binomial (Pascal) distribution with $r=5, p=1/2$:

$$P[X=10]=\binom{10-1}{5-1}\cdot\frac{1}{2^{10}}=63/512=0.123$$

(ii) Given 10 heads in 25 tosses, the first 24 tosses contain exactly 9 heads. Let X_r be the toss number of the r^{th} head ($1\le r\le 9$). To have the r^{th} head at toss x we need $r-1$ heads among tosses $1,\ldots,x-1$ and $9-r$ heads among tosses $x+1,\ldots,24$, so

$$\Pr(X_r=x)=\frac{\binom{x-1}{r-1}\binom{24-x}{9-r}}{\binom{24}{9}},\qquad r\le x\le 25-10+r.$$

Define $R_r(x)=\Pr(X_r=x+1)/\Pr(X_r=x)=\dfrac{x}{x-r+1}\cdot\dfrac{25-10+r-x}{24-x}$. The maximum is at the last x with $R_r(x)\ge 1$ (and also $x+1$ if $R_r(x)=1$).

5^{th} head ($r=5$). Setting $R_5(x)\ge 1$ and expanding, $x(25-10+5-x)\ge(x-4)(24-x)$, so $8x\le 96$ and $x\le 12$.

$$\Pr(X_5=12)=\Pr(X_5=13)=\frac{\binom{11}{4}\binom{12}{4}}{\binom{24}{9}}=\frac{330\cdot 495}{1307504}\approx 0.125.$$

The 5^{th} head is most likely on toss 12 or 13.

8^{th} head ($r = 8$). Setting $R_8(x) \geq 1$ gives $8x \leq 168$, so $x \leq 21$, and

$$\Pr(X_8 = 21) = \Pr(X_8 = 22) = \frac{\binom{20}{7}\binom{3}{1}}{\binom{24}{9}} = \frac{77520 \cdot 3}{1307504} \approx 0.178.$$

The 8^{th} head is most likely on toss 21 or 22.

(iii) For general m^{th} head on toss n ($m > 2$), $R_r(x) \geq 1$ is equivalent to

$$x(n - m + r - x) \geq (x - r + 1)(n - 1 - x).$$

The x^2 terms cancel, yielding the linear relationship

$$x \leq \frac{(r-1)(n-1)}{m-2}.$$

For $n = 25$, $m = 10$ this is $x = 3(r-1)$: $r = 5 \Rightarrow 12$, $r = 8 \Rightarrow 21$. Each unit increase in r has tied modules $\{3(r-1),\ 3(r-1)+1\}$, linear in r with slope 3.

The naive guess $\text{mode} \approx nr/m$ is inappropriate because it ignores the constraint that the m^{th} head is fixed at toss n. Linearity holds exactly, but with slope $(n-1)/(m-2)$, not n/m.

EX4. You flip a fair coin again. This time, you are interested in the average number of flips to get the first head in each round. You take the average of the n independent rounds.

(i) Let X_i be the number of flips in round i when the first head appears. What is the typical (long-run average) number of flips per round? Prove it by summing a series.

(ii) For $n = 10$, what is the probability this average is at most 1.5?

(iii) Let S_n be the total number of flips over all n rounds. Explain why S_n is the same flip index of the n^{th} head in one long fair-coin sequence as EX3, and prove that its probability generating function is

$$\sum_{x=n}^{\infty} \Pr(S_n = x)\, z^x = \frac{z^n}{(1-z)^n} \qquad (|z| < 1).$$

Hint: professor talked about counting in the journey of negative binomial distributi

(iv) Set $z = \frac{1}{2}$ in (iii). What does the identity say about the total probability?

(i) Each round is $\text{Geometric}(p = \frac{1}{2})$, so $\mathbb{E}[X_i] = 1/p = 2$. To prove this by a series,

$$\Pr(X_i = k) = \left(\frac{1}{2}\right)^k, \qquad k = 1, 2, \ldots, \qquad \mathbb{E}[X_i] = \sum_{k=1}^{\infty} k\left(\frac{1}{2}\right)^k := S.$$

$\frac{S}{2} = \sum_{k=1}^{\infty} k\left(\frac{1}{2}\right)^{k+1} = \sum_{j=2}^{\infty}(j-1)\left(\frac{1}{2}\right)^j$,

$S - \frac{S}{2} = \frac{1}{2} + \sum_{j=2}^{\infty}\left(\frac{1}{2}\right)^j = \frac{1}{2} + \frac{(1/2)^2}{1-1/2} = 1$. So $S = 2$.

(ii) An average ≤ 1.5 means $S_{10} = \sum_{i=1}^{10} X_i \leq 15$. As the waiting time for the 10^{th} head,

$$\Pr(S_{10}=x)=\binom{x-1}{9}\frac{1}{2^x}, \qquad x\ge 10.$$

Summing the six values $x=10,\dots,15$ gives

$$\Pr(S_{10}\le 15)=\frac{309}{2048}\approx 0.151.$$

(iii) In one infinite fair-coin sequence, "n rounds with total x flips" is the same event as "the n^{th} head occurs on flip x". The last flip of round n is that head, and the earlier $x-n$ flips contain exactly $n-1$ heads.

$$\Pr(S_n=x)=\binom{x-1}{n-1}\frac{1}{2^x}, \qquad x\ge n.$$

Multiply the PMF by z^x and sum. With $j=x-n$,

$$\sum_{x=n}^{\infty}\binom{x-1}{n-1}z^x=z^n\sum_{j=0}^{\infty}\binom{n+j-1}{n-1}z^j=\frac{z^n}{(1-z)^n},$$

using the negative-binomial series $\sum_{j=0}^{\infty}\binom{n+j-1}{j}r^j=(1-r)^{-n}$ with $r=z$.

(iv) At $z=\frac{1}{2}$, $\sum_{x=n}^{\infty}\Pr(S_n=x)=\frac{(1/2)^n}{(1-1/2)^n}=1$,

so the probability generating function encodes normalization of the tail distribution of S_n.

EX5. You train a neural network overnight as you sleep. But your local machine may fail halfway and disrupt the training. You hope the failure time occurs after most epochs are completed. We model the question as a widget with mean time between failures of 24 hours. It is working at 11 pm, left unattended, and found failed at 7 am next morning. Find the probability it was still working at 6 am.

(i) Model the lifetime as exponential with mean 24 h (constant failure rate).

(ii) A constant hazard rate may suit random failures but not wear-out. To model the same widget with increasing hazard, use a Weibull lifetime with shape $\beta=2$, and choose the scale α so the mean remains 24 h. (When $\beta=1$, this reduces to the exponential model in (i).) Repeat the calculation.

(iii) Does the exponential model have the memoryless property? Can a general Weibull model be memoryless? Prove your answer.

(i) $T\sim\text{Exponential}(\lambda=1/24)$. "Still working at 6 am" means $T>7$ h; "found failed by 7 am" means $T\le 8$ h.

$$\Pr(T>7\mid T\le 8)=\frac{\Pr(7<T\le 8)}{\Pr(T\le 8)}=\frac{e^{-7\lambda}-e^{-8\lambda}}{1-e^{-8\lambda}}=\frac{e^{-7/24}-e^{-1/3}}{1-e^{-1/3}}\approx 0.108.$$

(ii) For $T\sim\text{Weibull}(\alpha,\beta)$, $\mathbb{E}[T]=\alpha^{-1/\beta}\Gamma(1+1/\beta)$. With $\beta=2$ and $\mathbb{E}[T]=24$,

$$\alpha^{-1/2}\Gamma(3/2)=24 \quad\Rightarrow\quad \alpha=\left(\frac{\Gamma(3/2)}{24}\right)^2=\frac{\pi}{2304}\ \mathrm{h}^{-2}.$$

Reliability $R(t)=e^{-\alpha t^2}$. The same conditioning gives

$$\Pr(T>7\mid T\le 8)=\frac{R(7)-R(8)}{1-R(8)}=\frac{e^{-49\pi/2304}-e^{-64\pi/2304}}{1-e^{-64\pi/2304}}\approx 0.227.$$

Early in life the hazard $\rho(t) = 2\alpha t$ is below the exponential rate $1/24$, so wear-out modeling at equal mean makes survival up to hour 7 more likely once we know failure occurred within 8 hours.

(iii) Only the exponential model is memoryless.

T is memoryless if $\Pr(T > s + t \mid T > s) = \Pr(T > t)$ for all $s, t \geq 0$,

equivalently $R(s+t) = R(s)R(t)$ where $R(t) = \Pr(T > t)$.

Exponential. If $R(t) = e^{-\lambda t}$, then $R(s+t) = e^{-\lambda(s+t)} = e^{-\lambda s}e^{-\lambda t} = R(s)R(t)$.

Weibull. For $\beta \neq 1$, $R(t) = e^{-\alpha t^\beta}$ gives

$$\frac{R(s+t)}{R(s)} = \exp\big(-\alpha\big[(s+t)^\beta - s^\beta\big]\big), \qquad R(t) = \exp\left(-\alpha t^\beta\right).$$

Memorylessness requires $(s+t)^\beta - s^\beta = t^\beta$ for all $s, t > 0$. Divide by t^β and set $u = s/t$:

$$(u+1)^\beta - u^\beta = 1 \qquad \text{for all } u > 0.$$

Taking $u = 1$ gives $2^\beta = 2$, hence $\beta = 1$. Conversely, $\beta = 1$ gives $(u+1) - u = 1$, i.e. the exponential case. So the general Weibull family with $\beta \neq 1$ is not memoryless.

EX6. We consider a variant of EX1 (200 chips with defect rate $\theta = 0.03$), but we assume there are exactly 6 defectives. An inspector audits 30 chips without replacement.

(i) What is the probability the audit finds at most one defective chip?

(ii) Approximate the same probability using a binomial model with $p = 0.03$. Compare with (i).

(iii) Suppose instead the inspector audits half the lot (100 chips, still without replacement). Compare the hypergeometric answer to the binomial approximation again.

(iv) When does ignoring "without replacement" matter?

(i) Let X be the number of defectives in the audit. Then $X \sim \mathrm{Hyper}(n = 200,\ n_1 = 6,\ r = 30)$:

$$\Pr(X = k) = \frac{\binom{6}{k}\binom{194}{30-k}}{\binom{200}{30}}, \qquad k = 0, 1, \ldots, 6.$$

$$\Pr(X \leq 1) = \Pr(X = 0) + \Pr(X = 1) = \frac{\binom{194}{30}}{\binom{200}{30}} + \frac{6\binom{194}{29}}{\binom{200}{30}} \approx 0.778.$$

(ii) If sampling were with replacement, $X \approx \mathrm{Binomial}(30,\ 0.03)$:

$$\Pr(X \leq 1) = \binom{30}{0}(0.97)^{30} + \binom{30}{1}(0.03)(0.97)^{29} \approx 0.773.$$

The gap is small (≈ 0.005) as the audit takes only $30/200 = 15\%$ of the lot.

(iii) Now $X \sim \mathrm{Hyper}(N = 200,\ K = 6,\ r = 100)$.

$$\Pr(X=k)=\frac{\binom{6}{k}\binom{194}{100-k}}{\binom{200}{100}},\qquad k=0,1,\ldots,6.$$

$$\Pr(X\le 1)=\Pr(X=0)+\Pr(X=1)=\frac{\binom{194}{100}}{\binom{200}{100}}+\frac{6\binom{194}{99}}{\binom{200}{100}}\approx 0.0145+0.0914=0.106.$$

Binomial approximation. If each chip were defective independently with $p=K/N=0.03$, then $X\approx \mathrm{Binomial}(100,\,0.03)$

,

$$\Pr(X\le 1)=\binom{100}{0}(0.97)^{100}+\binom{100}{1}(0.03)(0.97)^{99}\approx 0.0484+0.1462=0.195.$$

The binomial value is almost twice the hypergeometric result (0.195 vs. 0.106).

(iv) The binomial approximation fails because: 1. there are only 6 defectives, so $X\le 6$ always, but the binomial model assigns positive probability to $X\ge 7$; 2. Once a defective is drawn, the remaining defectives in the population drop, so draws are negatively correlated. In particular,

$$\Pr_{\text{hyper}}(X=0)=\frac{\binom{194}{100}}{\binom{200}{100}}\approx 0.0145,\qquad \Pr_{\text{binom}}(X=0)=(0.97)^{100}\approx 0.0484.$$

The binomial approximation is reasonable when $f=r/N$ is small (rule of thumb $f\lesssim 10\%$) so that the finite-population correction factor $(N-r)/(N-1)\approx 1$ and exhausting the defective pool is unlikely. Equivalently, with $p=K/N$ fixed, require $r\ll N$; here $r=N/2$, so hypergeometric is necessary.

Part III. MGF

Geometric Distribution

Let $X\sim \mathrm{Geometric}(\theta)$, where X counts trials until the first success.
PMF: $P_X(k)=(1-\theta)^{k-1}\theta$ for $k=1,2,3,\ldots$

Derivation of MGF:

$$m_X(t)=\mathbb{E}(e^{tX})=\sum_{k=1}^{\infty}e^{tk}\cdot(1-\theta)^{k-1}\theta=\theta e^t\sum_{k=1}^{\infty}\left[(1-\theta)e^t\right]^{k-1}=\theta e^t\sum_{j=0}^{\infty}\left[(1-\theta)e^t\right]^j$$

This is a geometric series $\sum_{j=0}^{\infty}r^j=\frac{1}{1-r}$, valid when $|r|<1$.
Here $r=(1-\theta)e^t$, so convergence requires $(1-\theta)e^t<1$, i.e., $t<-\ln(1-\theta)$.

$$m_X(t)=\frac{\theta e^t}{1-(1-\theta)e^t},\quad t<-\ln(1-\theta)$$

First moment (mean):

$$m_X'(t)=\frac{\theta e^t\cdot[1-(1-\theta)e^t]-\theta e^t\cdot[-(1-\theta)e^t]}{[1-(1-\theta)e^t]^2}=\frac{\theta e^t}{[1-(1-\theta)e^t]^2}$$

Evaluate at $t=0$:

$$\mathbb{E}(X) = m_X'(0) = \frac{\theta}{[1-(1-\theta)]^2} = \frac{\theta}{\theta^2} = \frac{1}{\theta}$$

Second moment & variance:

$$m_X''(t) = \frac{\theta e^t\left[1+(1-\theta)e^t\right]}{[1-(1-\theta)e^t]^3}$$

$$\mathbb{E}(X^2) = m_X''(0) = \frac{\theta(1+1-\theta)}{\theta^3} = \frac{2-\theta}{\theta^2}$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \frac{2-\theta}{\theta^2} - \frac{1}{\theta^2} = \frac{1-\theta}{\theta^2}$$

Binomial Distribution

Let $X \sim \mathrm{Binomial}(n, \theta)$.
PMF: $P_X(k) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$ for $k = 0, 1, \dots, n$

Derivation of MGF:

$$m_X(t) = \mathbb{E}(e^{tX}) = \sum_{k=0}^{n} e^{tk}\binom{n}{k}\theta^k(1-\theta)^{n-k} = \sum_{k=0}^{n}\binom{n}{k}\left(\theta e^t\right)^k(1-\theta)^{n-k}$$

By the Binomial Theorem: $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k}$.
Set $a = \theta e^t$, $b = 1-\theta$:

$$m_X(t) = \left(1-\theta+\theta e^t\right)^n, \quad \forall t \in \mathbb{R}$$

Moments:

$$m_X'(t) = n\left(1-\theta+\theta e^t\right)^{n-1} \cdot \theta e^t$$

$$\mathbb{E}(X) = m_X'(0) = n(1)^{n-1}\theta = n\theta$$

$$m_X''(t) = n(n-1)\left(1-\theta+\theta e^t\right)^{n-2}(\theta e^t)^2 + n\left(1-\theta+\theta e^t\right)^{n-1}\theta e^t$$

$$\mathbb{E}(X^2) = n(n-1)\theta^2 + n\theta$$

$$\mathrm{Var}(X) = n(n-1)\theta^2 + n\theta - n^2\theta^2 = n\theta(1-\theta)$$

Negative Binomial Distribution

Let $X \sim \mathrm{NegBinomial}(r, \theta)$, where X = trial number of the r-th success.
PMF: $P_X(x) = \binom{x-1}{r-1}\theta^r(1-\theta)^{x-r}$ for $x = r, r+1, r+2, \dots$

Derivation of MGF: Let $Y = X - r$ (failures before the r-th success). Then

$$m_X(t) = e^{tr}\mathbb{E}(e^{tY}) = e^{tr}\theta^r \sum_{k=0}^{\infty}\binom{r+k-1}{k}\left[(1-\theta)e^t\right]^k$$

Use the generalized binomial theorem: $(1-x)^{-r} = \sum_{k=0}^{\infty}\binom{r+k-1}{k}x^k$ for $|x| < 1$.
Set $x = (1-\theta)e^t$, valid when $(1-\theta)e^t < 1$:

$$m_X(t) = \left(\frac{\theta e^t}{1-(1-\theta)e^t} \right)^r, \quad t < -\ln(1-\theta)$$

Moments:

X is the sum of r independent Geometric(θ) trial-count variables, so

$$\mathbb{E}(X) = \frac{r}{\theta}, \quad \mathrm{Var}(X) = \frac{r(1-\theta)}{\theta^2}$$

Poisson Distribution

Let $X \sim \mathrm{Poisson}(\lambda)$.
PMF: $P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}$ for $k = 0, 1, 2, \ldots$

Derivation of MGF:

$$m_X(t) = \mathbb{E}(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} \cdot \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!}$$

Recognize the Taylor series $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ with $x = \lambda e^t$:

$$m_X(t) = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)}, \quad \forall t \in \mathbb{R}$$

Moments:

$$m_X'(t) = e^{\lambda(e^t-1)} \cdot \lambda e^t$$

$$\mathbb{E}(X) = m_X'(0) = e^0 \cdot \lambda = \lambda$$

$$m_X''(t) = \lambda e^t \cdot e^{\lambda(e^t-1)} \cdot \lambda e^t + \lambda e^t \cdot e^{\lambda(e^t-1)}$$

$$\mathbb{E}(X^2) = \lambda^2 + \lambda$$

$$\mathrm{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Exponential Distribution

Let $X \sim \mathrm{Exponential}(\beta)$ (scale parameter β, rate $1/\beta$).
PDF: $f_X(x) = \frac{1}{\beta} e^{-x/\beta}$ for $x > 0$

Derivation of MGF:

$$m_X(t) = \mathbb{E}(e^{tX}) = \int_0^{\infty} e^{tx} \cdot \frac{1}{\beta} e^{-x/\beta} dx = \frac{1}{\beta} \int_0^{\infty} e^{-(1/\beta - t)x} dx$$

Converges only when $1/\beta - t > 0$, i.e., $t < 1/\beta$.

$$m_X(t) = \frac{1}{\beta} \cdot \frac{1}{1/\beta - t} = \frac{1}{1-\beta t} = (1-\beta t)^{-1}, \quad t < 1/\beta$$

Moments:

$$m_X'(t) = \beta(1-\beta t)^{-2} \implies \mathbb{E}(X) = \beta$$

$$m''_X(t) = 2\beta^2(1-\beta t)^{-3} \implies \mathbb{E}(X^2) = 2\beta^2$$
$$\text{Var}(X) = 2\beta^2 - \beta^2 = \beta^2$$

Gamma Distribution

Let $X \sim \text{Gamma}(\alpha, \beta)$.
PDF: $f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$ for $x > 0$

Derivation of MGF:

$$m_X(t) = \int_0^\infty e^{tx} \cdot \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} dx = \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty x^{\alpha-1} e^{-(1/\beta - t)x} dx$$

Let $u = (1/\beta - t)x$, so $x = \frac{u}{1/\beta - t}$, $dx = \frac{du}{1/\beta - t}$:

$$\begin{aligned} m_X(t) &= \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty \left(\frac{u}{1/\beta - t}\right)^{\alpha-1} e^{-u} \cdot \frac{du}{1/\beta - t} \\ &= \frac{1}{\Gamma(\alpha)\beta^\alpha (1/\beta - t)^\alpha} \int_0^\infty u^{\alpha-1} e^{-u} du \\ &= \frac{1}{\beta^\alpha (1/\beta - t)^\alpha} = \frac{1}{(1-\beta t)^\alpha} = (1-\beta t)^{-\alpha}, \quad t < 1/\beta \end{aligned}$$

Moments:

$$\mathbb{E}(X) = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2$$

Normal Distribution

Let $X \sim N(\mu, \sigma^2)$.
PDF: $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$

Derivation of MGF:

$$m_X(t) = \int_{-\infty}^\infty e^{tx} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx$$

Complete the square in the exponent:

$$\begin{aligned} tx - \frac{(x-\mu)^2}{2\sigma^2} &= -\frac{x^2 - 2\mu x + \mu^2 - 2\sigma^2 tx}{2\sigma^2} \\ &= -\frac{(x - (\mu + \sigma^2 t))^2 - (\mu + \sigma^2 t)^2 + \mu^2}{2\sigma^2} \\ &= -\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{1}{2}\sigma^2 t^2 \end{aligned}$$

Thus:

$$\begin{aligned} m_X(t) &= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-(\mu+\sigma^2 t))^2/(2\sigma^2)} dx \\ &= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot 1 \end{aligned}$$

The integral equals 1 because it is the total probability of a $N(\mu+\sigma^2 t, \sigma^2)$ density.

$$m_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}, \quad \forall t \in \mathbb{R}$$

MGF Toolkit

1. **Uniqueness Theorem**: If $m_X(t) = m_Y(t)$ for all t in some open neighborhood of 0, then X and Y have identical probability distributions.
2. **Independent Sums**: For independent $X_1, X_2, \ldots, X_n$:
   $m_{X_1+X_2+\cdots+X_n}(t) = \prod_{i=1}^n m_{X_i}(t)$
3. **Affine Transformation**: For constants a, b:
   $m_{aX+b}(t) = e^{bt} m_X(at)$
4. **Moment Extraction**: The k-th raw moment is the k-th derivative of the MGF evaluated at $t = 0$:
   $\mathbb{E}[X^k] = m_X^{(k)}(0)$
5. **Random Sum (Compound Distribution)**: If non-negative integer-valued N is independent of i.i.d. $X_1, X_2, \ldots$, and $S = \sum_{i=1}^N X_i$. Write $m_X(t) = \mathbb{E}[e^{tX}]$ and $m_N(s) = \mathbb{E}[e^{sN}]$:

   $$m_S(t) = \mathbb{E}\left[(m_X(t))^N\right] = m_N\left(\ln m_X(t)\right)$$

   Intuition. First fix a trial size X (MGF m_X), then let N randomize how many copies are added; the outer expectation is over N. Example: if $N \sim \mathrm{Poisson}(\lambda)$ and $X \sim \mathrm{Exp}(\beta)$, then $m_S(t) = m_N(\ln m_X(t)) = \exp\big(\lambda(\frac{1}{1-\beta t} - 1)\big)$.

   Proof. Condition on how many terms appear:

   $$m_S(t) = \mathbb{E}[e^{tS}] = \mathbb{E}\big[\mathbb{E}[e^{tS} \mid N]\big].$$

   Given $N = n$, the sum is $S = X_1 + \cdots + X_n$. Independence and identical distribution give

   $$\mathbb{E}[e^{tS} \mid N = n] = \mathbb{E}[e^{tX_1}]^n = \big(m_X(t)\big)^n.$$

   So $\mathbb{E}[e^{tS} \mid N] = \big(m_X(t)\big)^N$, and averaging over N yields

   $$m_S(t) = \mathbb{E}\big[\big(m_X(t)\big)^N\big].$$

   For the second form, note $\big(m_X(t)\big)^N = e^{N \ln m_X(t)}$. With $s = \ln m_X(t)$,

   $$\mathbb{E}\big[\big(m_X(t)\big)^N\big] = \mathbb{E}[e^{sN}] = m_N(s) = m_N\big(\ln m_X(t)\big).$$

EX1. Let $X_1 \sim \mathrm{Gamma}(\alpha_1, \beta)$ and $X_2 \sim \mathrm{Gamma}(\alpha_2, \beta)$ be independent with the same β.
Prove using MGFs that $X_1 + X_2 \sim \mathrm{Gamma}(\alpha_1 + \alpha_2, \beta)$.

Background: independent Gamma variables with a common scale β are closed under addition (e.g. the total waiting time until the $(\alpha_1 + \alpha_2)$-th event in a Poisson process is Erlang/Gamma).

Recall the Gamma MGF:
$m_X(t) = (1 - \beta t)^{-\alpha}, \quad t < 1/\beta$
By independence:
$m_{X_1+X_2}(t) = m_{X_1}(t) \cdot m_{X_2}(t) = (1-\beta t)^{-\alpha_1} \cdot (1-\beta t)^{-\alpha_2} = (1-\beta t)^{-(\alpha_1+\alpha_2)}$
This matches exactly the MGF of a $\mathrm{Gamma}(\alpha_1 + \alpha_2, \beta)$ distribution. By the uniqueness theorem,
$X_1 + X_2 \sim \mathrm{Gamma}(\alpha_1 + \alpha_2, \beta)$.

MGFs turn convolution integrals into simple multiplication.

EX2. Let $N \sim \text{Geometric}(\theta)$ (count of trials until first success, support $k = 1, 2, \ldots$) be independent of i.i.d. $X_i \sim \text{Exponential}(\beta)$. Define $S = \sum_{i=1}^{N} X_i$. Find the MGF of S and identify its exact distribution.

Background: this is a memoryless closure: summing a geometric (discrete memoryless) number of i.i.d. exponential (continuous memoryless) waiting times is again exponential, matching the total idle time until the first success in a Poisson-process model.

Apply the random sum MGF formula: $m_S(t) = m_N\left(\ln m_X(t)\right)$

- Geometric MGF: $m_N(s) = \frac{\theta e^s}{1-(1-\theta)e^s}$
- Exponential MGF: $m_X(t) = \frac{1}{1-\beta t}$

Substitute $e^s = m_X(t) = \frac{1}{1-\beta t}$:

$$m_S(t) = \frac{\theta \cdot \frac{1}{1-\beta t}}{1-(1-\theta)\cdot\frac{1}{1-\beta t}} = \frac{\theta}{(1-\beta t)-(1-\theta)} = \frac{1}{1-\frac{\beta}{\theta}t}$$

This is exactly the MGF of an Exponential distribution with scale parameter β/θ. By uniqueness, $S \sim \text{Exponential}(\beta/\theta)$.

EX3. Let $N \sim \text{Poisson}(\lambda)$. Each of the N events is independently labeled "Type A" with probability p and "Type B" with probability $1-p$. Let X count Type A events. Prove via MGFs that $X \sim \text{Poisson}(\lambda p)$.

Background: this is Poisson thinning: if each event in a Poisson(λ) stream is kept independently with probability p, the kept events form a Poisson(λp) process.

Given $N = n$, $X \sim \text{Binomial}(n, p)$ with conditional MGF $(1-p+pe^t)^n$. By law of total expectation:

$$m_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\mathbb{E}[e^{tX} \mid N]\right] = \mathbb{E}\left[(1-p+pe^t)^N\right]$$

This is exactly $m_N\left(\ln\left(1-p+pe^t\right)\right)$. The Poisson MGF is $m_N(s) = e^{\lambda(e^s-1)}$.

Substitute $e^s = 1-p+pe^t$: $m_X(t) = e^{\lambda((1-p+pe^t)-1)} = e^{\lambda p(e^t-1)}$.

This is the MGF of $\text{Poisson}(\lambda p)$. By uniqueness, $X \sim \text{Poisson}(\lambda p)$.

EX4. Let X, Y be i.i.d. with mean 0 and variance 1. Suppose $\frac{X+Y}{\sqrt{2}} \stackrel{d}{=} X$. Prove using MGFs that X must be standard normal.

Background: among distributions with finite variance, the normal is the only non-degenerate law stable under independent sums and rescaling: if averaging two i.i.d.\ copies recreates the same shape, that shape must be $N(0,1)$ (a finite-variance case of the CLT/stability principle).

Let $m(t)$ be the common MGF. The MGF of the normalized sum is: $m_{\frac{X+Y}{\sqrt{2}}}(t) = \mathbb{E}\left[e^{\frac{t}{\sqrt{2}}X}\right]\mathbb{E}\left[e^{\frac{t}{\sqrt{2}}Y}\right] = \left[m\left(\frac{t}{\sqrt{2}}\right)\right]^2$.

By distributional equality ($X \stackrel{d}{=} Y$ means X and Y follow the same distribution):

$m(t) = \left[m\left(\frac{t}{\sqrt{2}}\right)\right]^2$. Iterate n times: $m(t) = \left[m\left(\frac{t}{2^{n/2}}\right)\right]^{2^n}$.

Use the Taylor expansion $m(t) = 1 + \frac{t^2}{2} + o(t^2)$ (since mean=0, variance=1): $m(t) = \left[1 + \frac{t^2}{2 \cdot 2^n} + o\left(\frac{t^2}{2^n}\right)\right]^{2^n}$.

Take $n \to \infty$. Using $\lim_{n\to\infty}(1 + \frac{a}{n})^n = e^a$: $m(t) = e^{t^2/2}$.

This is the standard normal MGF. By uniqueness, $X \sim N(0, 1)$.

EX5. Let $\Lambda \sim \mathrm{Gamma}(\alpha, \beta)$. Given $\Lambda = \lambda$, let $X \sim \mathrm{Poisson}(\lambda)$. Derive the marginal MGF of X and show it matches a Negative Binomial distribution.

Background: this is a Gamma–Poisson mixture (Poisson rate with a Gamma prior); integrating out the random intensity produces overdispersed count data, and the marginal is negative binomial rather than Poisson.

By the law of total expectation,

$$m_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\big[\mathbb{E}[e^{tX} \mid \Lambda]\big] = \mathbb{E}\left[e^{\Lambda(e^t-1)}\right] = m_\Lambda(e^t - 1).$$

From Part III, $m_\Lambda(s) = (1 - \beta s)^{-\alpha}$ for $s < 1/\beta$. With $s = e^t - 1$,

$$m_X(t) = \left[1 - \beta(e^t - 1)\right]^{-\alpha} = \left[(1+\beta) - \beta e^t\right]^{-\alpha}, \qquad e^t - 1 < \frac{1}{\beta} \Leftrightarrow t < \ln\left(1 + \frac{1}{\beta}\right).$$

Factor out $(1+\beta)$: $m_X(t) = (1+\beta)^{-\alpha}\left(1 - \frac{\beta}{1+\beta}\, e^t\right)^{-\alpha}$.

Set $p = \dfrac{1}{1+\beta}$ (so $\dfrac{\beta}{1+\beta} = 1 - p$). Then $m_X(t) = \left(\frac{p}{1-(1-p)e^t}\right)^{\alpha}$.

This is the MGF of the failure-count negative binomial: if Y is the number of failures before the α-th success in i.i.d. $\mathrm{Bernoulli}(p)$ trials, Part III gives $\mathbb{E}[e^{tY}] = \big(p/(1-(1-p)e^t)\big)^{\alpha}$. Hence

$$\Pr(X = k) = \binom{k+\alpha-1}{\alpha-1} p^\alpha (1-p)^k, \qquad k = 0, 1, 2, \ldots,$$

with $\mathbb{E}[X] = \alpha(1-p)/p = \alpha\beta$ and $\mathrm{Var}(X) = \alpha(1-p)/p^2 = \alpha\beta(1+\beta)$.

This is not the trial-index $\mathrm{NegBinomial}(r, \theta)$ from Part III (which has support $\{r, r+1, \ldots\}$ and mean r/θ). Here X counts Poisson events under random rate Λ, matching the failure-count parametrization with $r = \alpha$ and $p = 1/(1+\beta)$.

Part IV. One Useful Application

A course has 27 lectures and exactly 5 attendance checks. Each time you choose to attend or skip before knowing whether there is an attendance check today. A successful showup is an attendance on a checked lecture. We want to optimize showup hit rate

$$R = \frac{\text{successful showups}}{\text{total showups}} = \frac{H}{S},$$

where H counts checked lectures you attend and S counts all lectures you attend ($S > 0$).

Compare four policies under two check models:

- Known schedule: evaluate on past term data, where checks take place at lectures $2,\ 9,\ 13,\ 20,\ 25$.
- Random schedule: the 5 checked lectures are a uniformly random subset of $\{1,\dots,27\}$, fixed for the term but unknown to you.

Policies:

1. Always attend.
2. Independent: attend each lecture independently with probability $p = 5/27$.
3. Check-is-due: at lecture i, let q_i be the number of consecutive unchecked lectures in $1,\dots,i-1$ (if none yet, count from lecture 1). Attend iff $q_i \geq 5$.
4. Budget: let c be the number of checks already observed in $1,\dots,i-1$. Attend iff $(5-c)/(27-i+1) \geq 5/27$ (equivalently $5i \geq 27c+5$).

(i) On the random schedule, find $\Pr\left(\text{lecture } 10 \text{ is checked} \mid \text{no checks in } 1,\dots,9\right)$.

(ii) For the random schedule, if each lecture is an independent check with probability $\lambda = 5/27$, write the lecture index T of the 5^{th} check as a sum of geometric waiting times and find $\mathbb{E}[T]$. What distribution would T follow if the arrivals were in continuous time?

(iii) On the known schedule, find H, S, R for policies $1,3,4$. Find $\mathbb{E}[H]$, $\mathbb{E}[S]$, and $\mathbb{E}[R]$ for policy 2.

(iv) On the random schedule, find $\mathbb{E}[H]$, $\mathbb{E}[S]$, and $\mathbb{E}[R]$ for all policies.

(i) Given no checks in $1,\dots,9$, all 5 checks lie in $10,\dots,27$ (18 slots), so $\Pr\left(\text{lecture } 10 \text{ checked}\right) = 5/18 \approx 0.278$.

(ii) Let X_j be the number of lectures from the $(j-1)^{\text{st}}$ check through the j^{th} check ($X_j \geq 1$, i.i.d. $\text{Geometric}(\lambda = 5/27)$ in the Bernoulli-per-lecture model). Then $T = \sum_{j=1}^{5} X_j$ and $\mathbb{E}[T] = 5/\lambda = 27$.

In continuous time, the waiting time to the 5^{th} event at rate λ is $\text{Gamma}(5,\ 1/\lambda)$.

(iii) Known schedule (checks at $2, 9, 13, 20, 25$).

Policy	S	H	R
Always attend	27	5	$5/27 \approx 0.185$
Check-is-due	4 (8,9,19,20)	2 (9,20)	$1/2$
Budget	13	5	$5/13 \approx 0.385$
Independent	$\mathbb{E}[S] = 5$	$\mathbb{E}[H] = 25/27$	$\mathbb{E}[R] = 5/27$

Check-is-due attends $8, 9, 19, 20$ (wasted show-ups at $8, 19$). Budget attends at lectures $1, 2, 7, 8, 9, 12, 13, 18, 19, 20, 23, 24, 25$ and hits all five checks.

For independent policy, $H \sim \text{Binomial}(5,\ p)$ and $S \sim \text{Binomial}(27,\ p)$ with $p = 5/27$, so $\mathbb{E}[H] = 25/27$ and $\mathbb{E}[S] = 5$; given $S = s > 0$, each show-up hits with probability $5/27$, hence $\mathbb{E}[R] = 5/27$.

(iv) **Random schedule.** There are $\binom{27}{5} = 80{,}730$ equally likely check sets C. Write A for the showup set determined by the policy. Linearity gives $\mathbb{E}[H] = \sum_i \Pr(H_i)$ and $\mathbb{E}[S] = \sum_i \Pr(A_i)$, but $R = H/S$ is nonlinear, so we need

$$\mathbb{E}[R] = \sum_{s\geq 1} \Pr(S=s)\,\frac{\mathbb{E}[H \mid S=s]}{s} = \frac{1}{\binom{27}{5}} \sum_C \frac{|A\cap C|}{|A|}.$$

In general $\mathbb{E}[R] \neq \mathbb{E}[H]/\mathbb{E}[S]$ (ratio of expectations is not the expectation of a ratio).

Policy	$\mathbb{E}[H]$	$\mathbb{E}[S]$	$\mathbb{E}[R]$
Always attend	5	27	$5/27 \approx 0.185$
Independent	$25/27$	5	$5/27$
Check-is-due	$1463/897 \approx 1.63$	$32186/4485 \approx 7.18$	$1951580767/7326839520 \approx 0.266$
Budget	$6607/1794 \approx 3.68$	14	$5935029173/17509326120 \approx 0.339$

Always attend. Every check set has $|C| = 5$, and the policy attends all 27 lectures, so $H = 5$ and $S = 27$ always. Hence $\mathbb{E}[H] = 5$, $\mathbb{E}[S] = 27$, and $\mathbb{E}[R] = H/S = 5/27$.

Independent. Attend each lecture with $p = 5/27$, independently of C. Then $H \sim \mathrm{Binomial}(5,p)$ (each of the five checks is hit independently) and $S \sim \mathrm{Binomial}(27,p)$, so $\mathbb{E}[H] = 5p = 25/27$ and $\mathbb{E}[S] = 27p = 5$. Given $S = s > 0$, each attended lecture is a check with probability p, so $\mathbb{E}[H \mid S=s] = sp$ and $\mathbb{E}[R \mid S=s] = p$. Averaging over S (or applying $\mathbb{E}[R] = \mathbb{E}[\mathbb{E}[R \mid S]]$) gives $\mathbb{E}[R] = 5/27$.

Check-is-due. Fix $i \geq 6$. Attend at i iff $q_i \geq 5$, i.e.\ no check in $1,\dots,i-1$ or the last check $j \in \{1,\dots,i-6\}$ (then $i-1-j \geq 5$ unchecked lectures after j). Count favorable 5-subsets C:

- **No check before i.** All five checks lie in $\{i,\dots,27\}$: $\binom{27-i+1}{5} = \binom{28-i}{5}$.
- **Last check at $j \leq i-6$.** Place a check at j, none in $j+1,\dots,i-1$, and distribute the other four checks among the $(j-1)+(28-i)$ allowed positions $\{1,\dots,j-1\} \cup \{i,\dots,27\}$: $\binom{(j-1)+(28-i)}{4}$. With $k = j-1$,

$$\sum_{j=1}^{i-6} \binom{(j-1)+(28-i)}{4} = \sum_{k=0}^{i-7} \binom{28-i+k}{4}.$$

Hockey-stick / Pascal collapse. Use

$$\sum_{k=0}^{m} \binom{a+k}{a} = \binom{a+m+1}{a+1} \qquad \text{(hockey-stick identity)}$$

and Pascal's rule $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$. For every $i = 6,\dots,27$,

$$\binom{28-i}{5} + \sum_{k=0}^{i-7} \binom{28-i+k}{4} = \binom{22}{5},$$

because the second sum telescopes with the first term via $m = i-7$ repeated applications of Pascal (equivalently, one hockey-stick plus one Pascal step closes at $\binom{22}{5}$). Thus $\Pr(A_i) = \binom{22}{5}/\binom{27}{5}$ is constant on $i = 6,\dots,27$; lectures $1,\dots,5$ never qualify ($q_i \leq 4$).

For $i = 9$, the four-term expansion is

$$\binom{19}{5} + \binom{19}{4} + \binom{20}{4} + \binom{21}{4} = \binom{22}{5} = 26334,$$

using $\binom{19}{5}+\binom{19}{4}=\binom{20}{5}$, then $\binom{20}{5}+\binom{20}{4}=\binom{21}{5}$, then $\binom{21}{5}+\binom{21}{4}=\binom{22}{5}$ (three Pascal steps). Hence

$$\Pr(A_9)=\frac{26334}{80730}=\frac{1463}{4485}\approx 0.326.$$

Therefore

$$\mathbb{E}[S]=22\cdot\frac{\binom{22}{5}}{\binom{27}{5}}=22\cdot\frac{1463}{4485}=\frac{32186}{4485}\approx 7.18.$$

Hits. For $\Pr(H_i)=\Pr(i\in C,\ A_i)$, also require $i\in C$. Case on the last check j in $1,\dots,i-1$ (take $j=0$ if none):

j	**Pattern of checks in** $1,\dots,i-1$	**Count**
0	none	$\binom{27-i}{4}$
1	$\{1\}$ only	$\binom{27-i}{3}$
2	$\{2\}$ or $\{1,2\}$	$\binom{27-i}{3}+\binom{27-i}{2}$
3	$\{3\}$, $\{1,3\}$, $\{2,3\}$, or $\{1,2,3\}$	$\binom{27-i}{3}+2\binom{27-i}{2}+\binom{27-i}{1}$

At $i=9$ ($27-i=18$): $3060+816+969+1140=5985$, so

$$\Pr(H_9)=\frac{5985}{80730}=\frac{133}{1794},\qquad \mathbb{E}[H]=22\cdot\frac{133}{1794}=\frac{1463}{897}\approx 1.63$$

(same value for every $i=6,\dots,27$ by symmetry of the random check set).

$\mathbb{E}[R]$ **(check-is-due).** Use the tower property with $S=|A|$:

$$\mathbb{E}[R]=\sum_{s\ge 0}\Pr(S=s)\,\mathbb{E}\left[\frac{H}{S}\;\middle|\;S=s\right]=\sum_{s\ge 1}\Pr(S=s)\,\frac{\mathbb{E}[H\mid S=s]}{s}.$$

Rows $(s,\ \Pr(S=s),\ \mathbb{E}[H\mid S=s],\ \text{contrib})$ with $\text{contrib}=\Pr(S=s)\cdot\mathbb{E}[H\mid S=s]/s$:

$$(0,\tfrac{28}{40365},0,0),\ (1,\tfrac{14}{2691},\tfrac{5}{6},\tfrac{35}{8073}),\ (2,\tfrac{13}{690},\tfrac{1930}{1521},\tfrac{193}{16146}),\ (3,\tfrac{121}{2691},\tfrac{365}{242},\tfrac{365}{16146}),\ (4,\tfrac{101}{1242},\tfrac{2155}{1313},\tfrac{2155}{64584}),$$
$$(5,\tfrac{964}{8073},\tfrac{825}{482},\tfrac{110}{2691}),\ (6,\tfrac{2333}{16146},\tfrac{4040}{2333},\tfrac{1010}{24219}),\ (7,\tfrac{6073}{40365},\tfrac{20825}{12146},\tfrac{595}{16146}),\ (8,\tfrac{1121}{8073},\tfrac{3793}{2242},\tfrac{3793}{129168}),\ (9,\tfrac{908}{8073},\tfrac{374}{227},\tfrac{1496}{72657}),$$
$$(10,\tfrac{1273}{16146},\tfrac{1996}{1273},\tfrac{499}{40365}),\ (11,\tfrac{401}{8073},\tfrac{1165}{802},\tfrac{1165}{177606}),\ (12,\tfrac{1193}{40365},\tfrac{3365}{2386},\tfrac{673}{193752}),\ (13,\tfrac{122}{8073},\tfrac{163}{122},\tfrac{163}{104949}),\ (14,\tfrac{103}{16146},\tfrac{122}{103},\tfrac{61}{113022}),$$
$$(15,\tfrac{19}{8073},\tfrac{35}{38},\tfrac{7}{48438}),\ (16,\tfrac{8}{8073},\tfrac{15}{16},\tfrac{5}{86112}),\ (17,\tfrac{1}{3105},\tfrac{25}{26},\tfrac{5}{274482}),\ (18,\tfrac{1}{16146},1,\tfrac{1}{290628}).$$

Summing contributions gives $\mathbb{E}[R]=1951580767/7326839520\approx 0.266$.

Budget. Let c = checks in $1,\dots,i-1$. Attend at i iff the remaining check rate is at least $5/27$:

$$\frac{5-c}{27-i+1}\ge\frac{5}{27}\quad\iff\quad 5i\ge 27c+5.$$

Choose c checks among $1,\dots,i-1$ and $5-c$ among $i,\dots,27$:

$$\Pr(A_i)=\frac{1}{\binom{27}{5}}\sum_{c:\,5i\ge 27c+5}\binom{i-1}{c}\binom{27-i+1}{5-c}.$$

Summing over i and exchanging the order of summation (each pair (C,i) with $i\in A(C)$ counted once) yields

$$\sum_{i=1}^{27} \Pr(A_i) = \frac{14\binom{27}{5}}{\binom{27}{5}} = 14, \qquad \text{so} \qquad \mathbb{E}[S] = 14.$$

For $i = 10$, only $c = 0, 1$ satisfy $50 \geq 27c + 5$:

$$\Pr(A_{10}) = \frac{\binom{9}{0}\binom{18}{5} + \binom{9}{1}\binom{18}{4}}{\binom{27}{5}} = \frac{8568 + 27540}{80730} = \frac{36108}{80730} \approx 0.447.$$

Hits (budget). Require $i \in C$ as well; place c checks before i (including i itself), and $4 - c$ after i:

$$\Pr(H_i) = \frac{1}{\binom{27}{5}} \sum_{c:\, 5i \geq 27c+5} \binom{i-1}{c}\binom{27-i}{4-c}.$$

In particular $\Pr(H_1) = \binom{26}{4}/\binom{27}{5} = 5/27$. Summing over i,

$$\mathbb{E}[H] = \sum_{i=1}^{27} \Pr(H_i) = \frac{6607}{1794} \approx 3.68.$$

$\mathbb{E}[R]$ **(budget).** The policy is label-equivariant: for any permutation π of $\{1, \ldots, 27\}$ and check set C, $|A(C)| = |A(\pi(C))|$ (relabel lectures and checks together). Because C is uniform on $\binom{27}{5}$, every showup count $s \in \{1, \ldots, 27\}$ arises from exactly 2990 check sets ($2990 \times 27 = 80730$), so $\Pr(S = s) = 1/27$.

For $s \geq 22$, the budget constraint forces $A(C) \supseteq C$ (all five checks are attended), hence $\mathbb{E}[H \mid S = s] = 5$ and $\mathbb{E}[H \mid S = s]/s = 5/s$. For $s \leq 21$, the conditional expectations below come from enumerating check sets with $|A(C)| = s$.

With $\mathbb{E}[R] = \frac{1}{27}\sum_{s=1}^{27} \mathbb{E}[H \mid S = s]/s$, rows $(s,\ \mathbb{E}[H \mid S = s],\ \mathbb{E}[H \mid S = s]/s)$:

$$(1, 1, 1),\ (2, \tfrac{775}{598}, \tfrac{775}{1196}),\ (3, \tfrac{476}{299}, \tfrac{476}{897}),\ (4, \tfrac{1129}{598}, \tfrac{1129}{2392}),\ (5, \tfrac{653}{299}, \tfrac{653}{1495}),\ (6, \tfrac{1483}{598}, \tfrac{1483}{3588}),\ (7, \tfrac{35}{13}, \tfrac{5}{13}),$$

$$(8, \tfrac{1737}{598}, \tfrac{1737}{4784}),\ (9, \tfrac{932}{299}, \tfrac{932}{2691}),\ (10, \tfrac{1991}{598}, \tfrac{1991}{5980}),\ (11, \tfrac{1059}{299}, \tfrac{1059}{3289}),\ (12, \tfrac{85}{23}, \tfrac{85}{276}),\ (13, \tfrac{1151}{299}, \tfrac{1151}{3887}),\ (14, \tfrac{1197}{299}, \tfrac{1197}{4186}),$$

$$(15, \tfrac{1243}{299}, \tfrac{1243}{4485}),\ (16, \tfrac{1289}{299}, \tfrac{1289}{4784}),\ (17, \tfrac{1335}{299}, \tfrac{1335}{5083}),\ (18, \tfrac{1367}{299}, \tfrac{1367}{5382}),\ (19, \tfrac{1399}{299}, \tfrac{1399}{5681}),\ (20, \tfrac{1431}{299}, \tfrac{1431}{5980}),\ (21, \tfrac{1463}{299}, \tfrac{1463}{6279}),$$

$$(22, 5, \tfrac{5}{22}),\ (23, 5, \tfrac{5}{23}),\ (24, 5, \tfrac{5}{24}),\ (25, 5, \tfrac{5}{25}),\ (26, 5, \tfrac{5}{26}),\ (27, 5, \tfrac{5}{27}).$$

Multiplying the last column by $1/27$ and summing gives $\mathbb{E}[R] = 5935029173/17509326120 \approx 0.339$.